Lecture 13: Linearity, Thévenin and Norton Equivalents

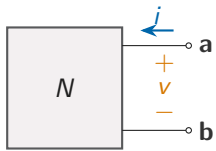
EE201 – Circuit Theory I

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Department of Electrical and Electronics Engineering
Middle East Technical University

current voltage, node voltage e Δ Y

Review: equivalent one-ports



- Equivalent: the same i–v characteristic
- Series, parallel, source transformation

Roadmap





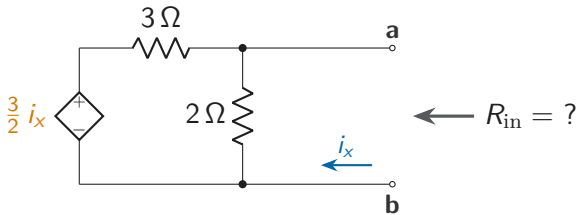






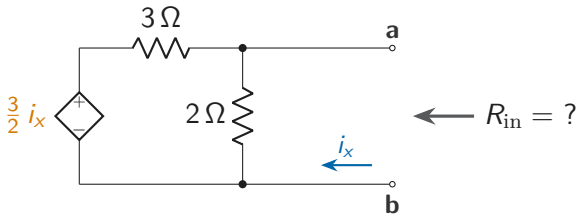
Input resistance	**Linearity**	**Superposition**	**Thévenin, Norton**	**Maximum power**
1	2	3	4	5
a test source **a dependent source**	$y = \mathbf{k}\,\mathbf{u}$ the implication	one at a time three experiments	two theorems open, short negative R_{th} two examples	the power P_L $R_L = R_{\text{th}}$ two examples

Motivation



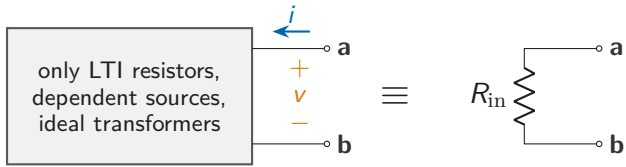
- A dependent source inside: series, parallel are not enough

Motivation

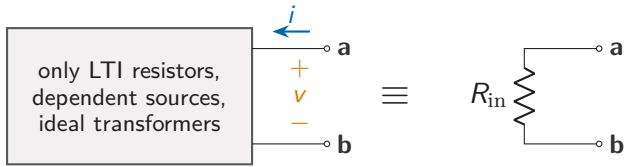


- A dependent source inside: series, parallel are not enough
- One equivalent resistor? Yes: the input resistance

One-ports without independent sources

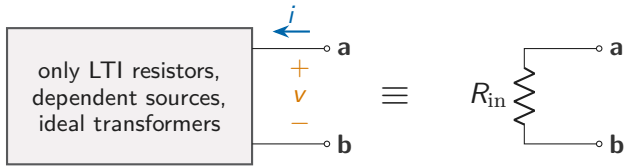


One-ports without independent sources



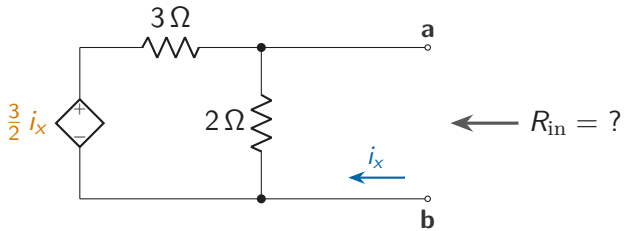
- Controls of the dependent sources: inside the one-port

One-ports without independent sources

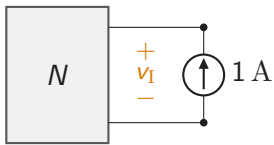


- Controls of the dependent sources: inside the one-port
- $v = R_{\text{in}}\, i$: R_{in} is the input resistance

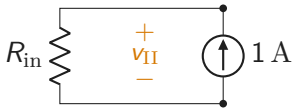
Example 13.1: an input resistance



The idea: a test source

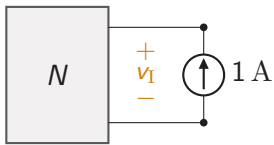


(I)

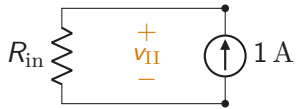


(II)

The idea: a test source



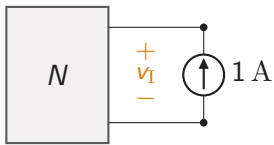
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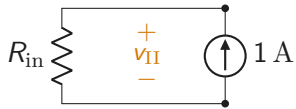
(II)

$$N \equiv R_{\mathrm{in}} \;\Rightarrow\; v_{\mathrm{I}} = v_{\mathrm{II}} = R_{\mathrm{in}} \cdot 1\,\mathrm{A}$$

The idea: a test source



(I)

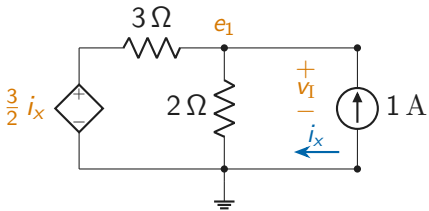


(II)

$$N \equiv R_{in} \;\Rightarrow\; v_I = v_{II} = R_{in} \cdot 1\,\mathrm{A}$$

$$R_{in} = \frac{v_I}{1\,\mathrm{A}}\text{: solve circuit (I)}$$

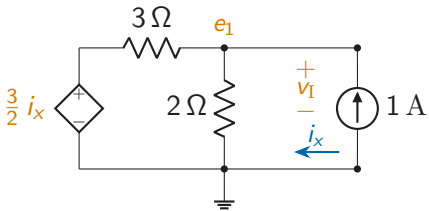
Example 13.1: the solution



Node 1:

$$\frac{e_1 - \frac{3}{2}\, i_x}{3} + \frac{e_1}{2} - 1 = 0$$

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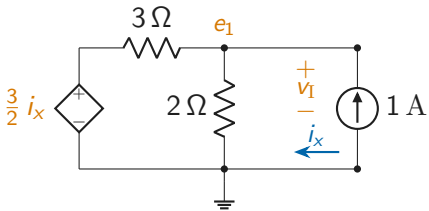


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$i_x = ?$

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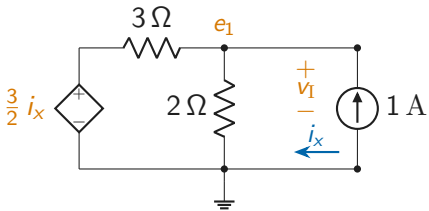
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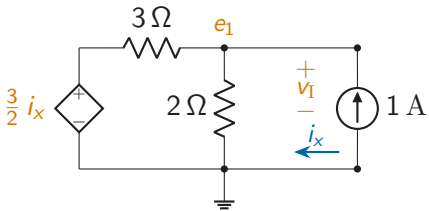
$$\frac{e_1 - \frac{3}{2}\, i_x}{3} + \frac{e_1}{2} - 1 = 0$$

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$i_x = -1\,\text{A}$

$$\frac{e_1 + \frac{3}{2}}{3} + \frac{e_1}{2} - 1 = 0 \;\Rightarrow\; e_1 = \tfrac{3}{5}\,\text{V}$$

Example 13.1: the solution



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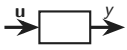
$i_x = -1\,\mathrm{A}$

$$\frac{e_1 + \frac{3}{2}}{3} + \frac{e_1}{2} - 1 = 0 \;\Rightarrow\; e_1 = \tfrac{3}{5}\,\mathrm{V}$$

$$R_\mathrm{in} = \frac{v_\mathrm{I}}{1\,\mathrm{A}} = \frac{3}{5}\,\Omega$$

Roadmap











Input resistance ✓

a test source ✓
a dependent source ✓

Linearity 2

$y = \mathbf{k}\,\mathbf{u}$
the implication

Superposition 3

one at a time
three experiments

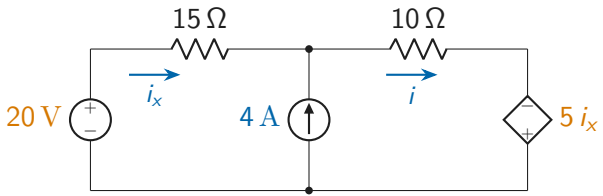
Thévenin, Norton 4

two theorems
open, short
negative R_{th}
two examples

Maximum power 5

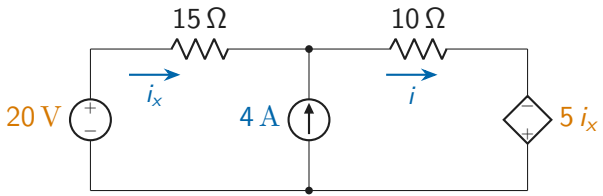
the power P_L
$R_L = R_{\text{th}}$
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Motivation: many sources



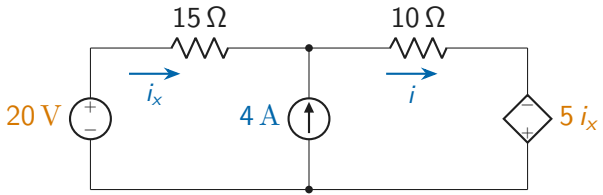
- Two independent sources: the inputs

Motivation: many sources



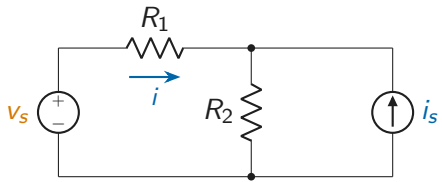
- Two independent sources: the inputs
- One source at a time: two simpler circuits?

Motivation: many sources



- Two independent sources: the inputs
- One source at a time: two simpler circuits?
- Yes: the circuit is linear

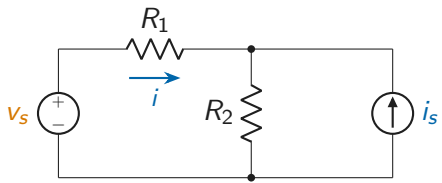
Linearity: inputs and outputs



Input vector: the independent sources

$$\mathbf{u} = \begin{bmatrix} v_s \\ i_s \end{bmatrix}$$

Linearity: inputs and outputs



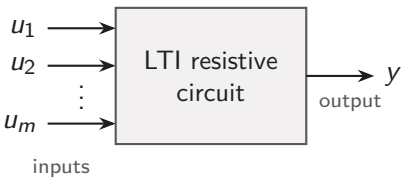
Input vector: the independent sources

$$\mathbf{u} = \begin{bmatrix} v_s \\ i_s \end{bmatrix}$$

Output y: any branch voltage or current

e.g. $y = i$

Linearity

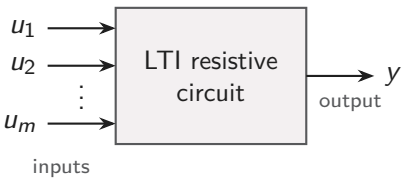


Input u_j alone (the others zero):

$y_j = k_j\, u_j$

k_j: a constant (resistors, gains)

Linearity



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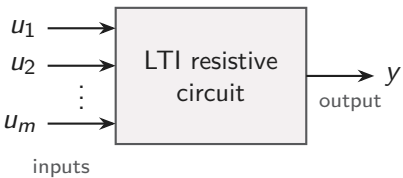
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All inputs: the sum

$y = y_1 + y_2 + \cdots + y_m$

Linearity



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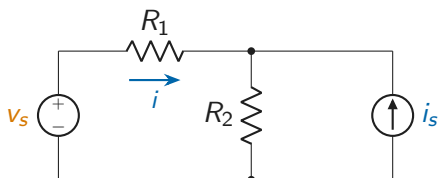
$y = y_1 + y_2 + \cdots + y_m$

$$y = k_1 u_1 + \cdots + k_m u_m = \mathbf{k}\,\mathbf{u}$$

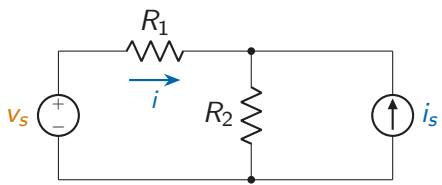
Example 13.2: k for $y = i$

$$i = i_1 + i_2, \quad i_1 = k_1 v_s, \quad i_2 = k_2 i_s$$

$$k_1, k_2 = ?$$



Example 13.2: k for $y = i$

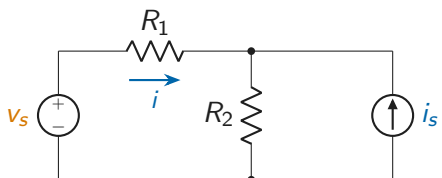


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KVL: $-v_s + R_1 i + R_2(i + i_s) = 0$

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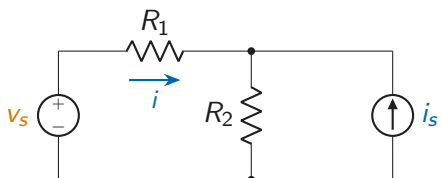
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$$i = \underbrace{\frac{1}{R_1 + R_2} v_s}_{i_1} \underbrace{- \frac{R_2}{R_1 + R_2} i_s}_{i_2}$$

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$$k_1 = \frac{1}{R_1 + R_2}, \qquad k_2 = -\frac{R_2}{R_1 + R_2}$$

Linearity: the implication

$$y(\alpha\,\mathbf{u} + \beta\,\mathbf{w}) = \alpha\,y(\mathbf{u}) + \beta\,y(\mathbf{w})$$

$\mathbf{u}$, $\mathbf{w}$: input vectors; α, β: numbers

Linearity: the implication

$$y(\alpha\,\mathbf{u} + \beta\,\mathbf{w}) = \alpha\,y(\mathbf{u}) + \beta\,y(\mathbf{w})$$

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$$y\left(\begin{bmatrix} v_s \\ i_s \end{bmatrix}\right) = y\left(\begin{bmatrix} v_s \\ 0 \end{bmatrix}\right) + y\left(\begin{bmatrix} 0 \\ i_s \end{bmatrix}\right)$$

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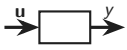
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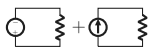
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Superposition: one independent source at a time

Roadmap











Input resistance ✓

a test source ✓
a dependent source ✓

Linearity ✓

$y = \mathbf{k}\,\mathbf{u}$ ✓
the implication ✓

Superposition 3

one at a time
three experiments

Thévenin, Norton 4

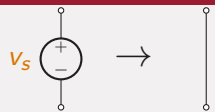
two theorems
open, short
negative R_{th}
two examples

Maximum power 5

the power P_L
$R_L = R_{\text{th}}$
two examples

Killing a source

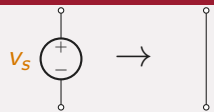
Voltage source



$v_s = 0$: a short circuit

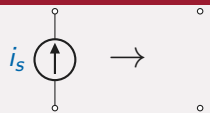
Killing a source

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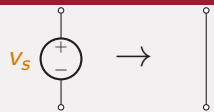
Current source



$i_s = 0$: an open circuit

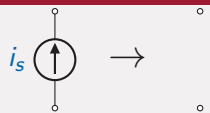
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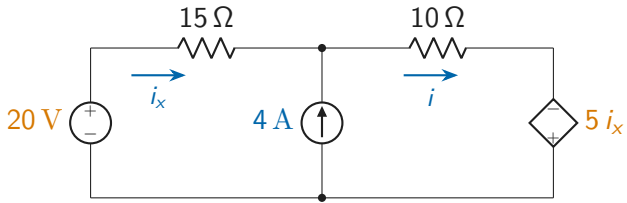
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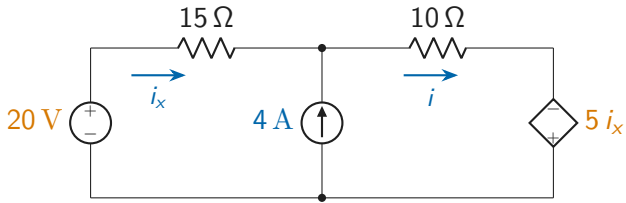
Do not kill dependent sources: they are not inputs

Example 13.3: superposition



Find i, one source at a time

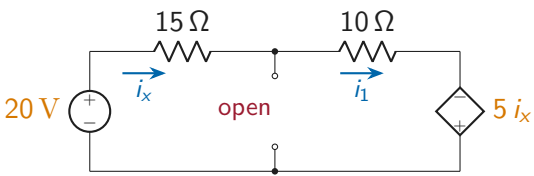
Example 13.3: superposition



Find i, one source at a time

$i = i_1 + i_2$: i_1: only 20 V, i_2: only 4 A

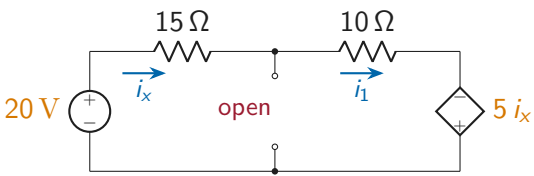
Example 13.3: kill the current source



KVL:

$$-20 + 15\,i_x + 10\,i_1 - 5\,i_x = 0$$

Example 13.3: kill the current source

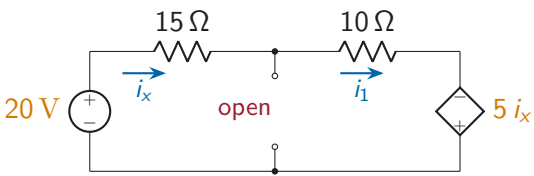


KVL:

$$-20 + 15\, i_x + 10\, i_1 - 5\, i_x = 0$$

$$i_x = ?$$

Example 13.3: kill the current source



KVL:

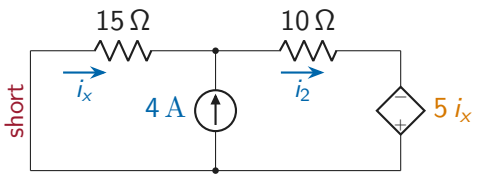
$$-20 + 15\, i_x + 10\, i_1 - 5\, i_x = 0$$

$i_x = ?$

$i_x = i_1 \;\Rightarrow\; 20\, i_1 = 20$

$$\boxed{i_1 = 1\,\text{A}}$$

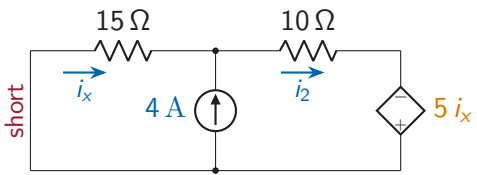
Example 13.3: kill the voltage source



KVL:

$$15\, i_x + 10\, i_2 - 5\, i_x = 0$$

Example 13.3: kill the voltage source

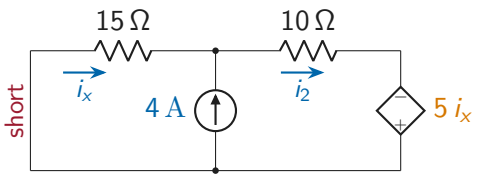


KVL:

$15\,i_x + 10\,i_2 - 5\,i_x = 0$

$i_x = ?$

Example 13.3: kill the voltage source



KVL:

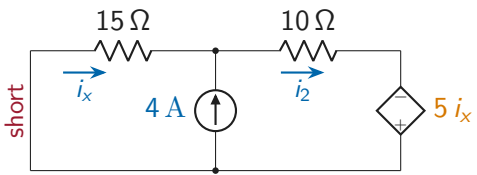
$15\,i_x + 10\,i_2 - 5\,i_x = 0$

$i_x = ?$

$i_x = i_2 - 4 \;\Rightarrow\; 20\,i_2 = 40$

$i_2 = 2\,\text{A}$

Example 13.3: kill the voltage source



KVL:

$15\,i_x + 10\,i_2 - 5\,i_x = 0$

$i_x = ?$

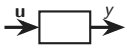
$i_x = i_2 - 4 \;\Rightarrow\; 20\,i_2 = 40$

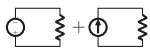
$i_2 = 2\,\mathrm{A}$

$i = i_1 + i_2 = 3\,\mathrm{A}$

Roadmap











Input resistance ✓

a test source ✓
a dependent source ✓

Linearity ✓

$y = \mathbf{k}\,\mathbf{u}$ ✓
the implication ✓

Superposition 3

one at a time ✓
three experiments

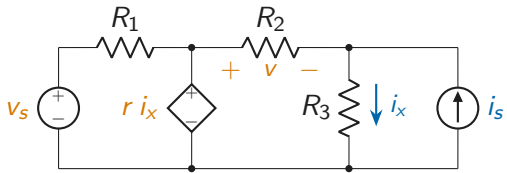
Thévenin, Norton 4

two theorems
open, short
negative R_{th}
two examples

Maximum power 5

the power P_L
$R_L = R_{\text{th}}$
two examples

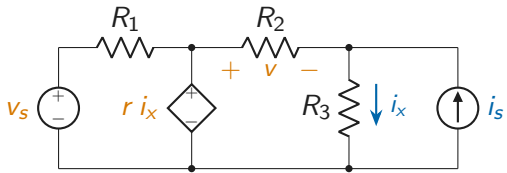
Example 13.4: three experiments



	v_s	i_s	v
Exp. 1	$12\,\mathrm{V}$	0	v_1
Exp. 2	$-6\,\mathrm{V}$	$6\,\mathrm{A}$	v_2
Exp. 3	$4\,\mathrm{V}$	$4\,\mathrm{A}$	v_3

v_3 in terms of v_1 and v_2?

Example 13.4: three experiments



	v_s	i_s	v
Exp. 1	$12\,\mathrm{V}$	0	v_1
Exp. 2	$-6\,\mathrm{V}$	$6\,\mathrm{A}$	v_2
Exp. 3	$4\,\mathrm{V}$	$4\,\mathrm{A}$	v_3

v_3 in terms of v_1 and v_2?

Linearity: $v = \mathbf{k} \begin{bmatrix} v_s \\ i_s \end{bmatrix}$

Example 13.4: the solution

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix}$$

Example 13.4: the solution

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix} \qquad \Rightarrow \qquad \beta = \tfrac{2}{3}, \quad \alpha = \tfrac{2}{3}$$

Example 13.4: the solution

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix} \qquad \Rightarrow \qquad \beta = \tfrac{2}{3}, \quad \alpha = \tfrac{2}{3}$$

$$v_3 = v\left(\alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix}\right) = \alpha\, v_1 + \beta\, v_2$$

Example 13.4: the solution

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix} \qquad \Rightarrow \qquad \beta = \tfrac{2}{3}, \quad \alpha = \tfrac{2}{3}$$

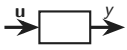
$$v_3 = v\left(\alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix}\right) = \alpha\, v_1 + \beta\, v_2$$

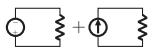
$$v_3 = \tfrac{2}{3}\, v_1 + \tfrac{2}{3}\, v_2$$

No element values needed

Roadmap











Input resistance ✓

a test source ✓
a dependent source ✓

Linearity ✓

$y = \mathbf{k}\,\mathbf{u}$ ✓
the implication ✓

Superposition ✓

one at a time ✓
three experiments ✓

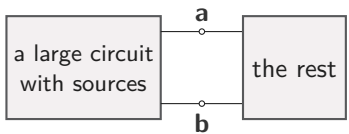
Thévenin, Norton 4

two theorems
open, short
negative R_{th}
two examples

Maximum power 5

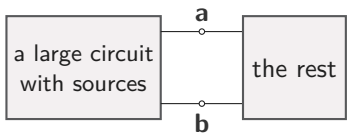
the power P_L
$R_L = R_{\text{th}}$
two examples

Motivation: a one-port with sources



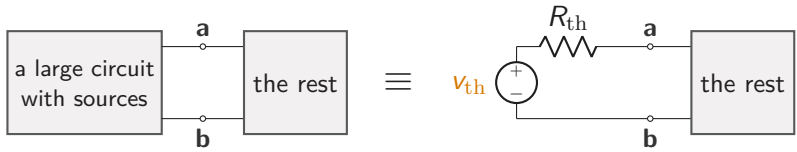
- The rest: a load (a lamp, a motor) or another circuit

Motivation: a one-port with sources



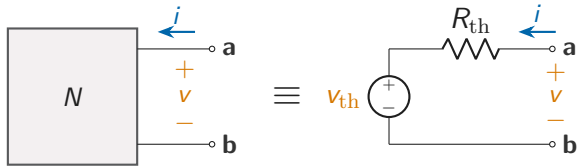
- The rest: a load (a lamp, a motor) or another circuit
- The rest changes: solve the large circuit again each time?

Motivation: a one-port with sources

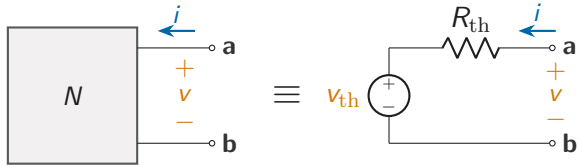


- The rest: a load (a lamp, a motor) or another circuit
- The rest changes: solve the large circuit again each time?
- No: the port sees one source and one resistor, found once

Thévenin's theorem

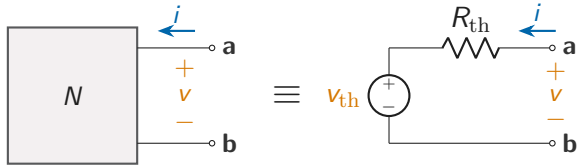


Thévenin's theorem



- N: LTI resistive, with independent sources
- (A1) Controls of the dependent sources: inside N
- (A2) The port is both voltage- and current-controlled

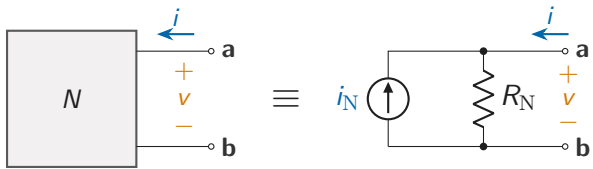
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- N: LTI resistive, with independent sources
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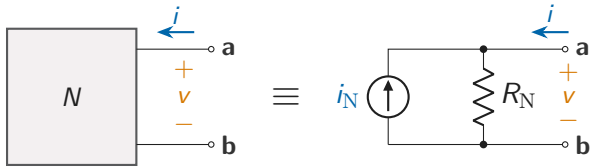
$$v = R_{\text{th}}\, i + v_{\text{th}}$$

Norton's theorem



Same conditions (A1), (A2)

Norton's theorem



Same conditions (A1), (A2)

$$v = R_N\, i + R_N\, i_N$$

Thévenin and Norton: the same line

$$v = R_{\mathrm{th}}\, i + v_{\mathrm{th}}$$

$$v = R_{\mathrm{N}}\, i + R_{\mathrm{N}}\, i_{\mathrm{N}}$$

Thévenin and Norton: the same line

$$v = R_{\mathrm{th}}\, i + v_{\mathrm{th}}$$

$$v = R_{\mathrm{N}}\, i + R_{\mathrm{N}}\, i_{\mathrm{N}}$$

$$R_{\mathrm{th}} = R_{\mathrm{N}}$$
$$v_{\mathrm{th}} = R_{\mathrm{th}}\, i_{\mathrm{N}}$$

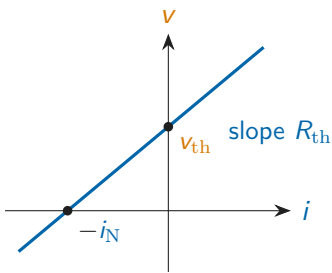
Thévenin and Norton: the same line

$v = R_{\text{th}}\, i + v_{\text{th}}$

$v = R_{\text{N}}\, i + R_{\text{N}}\, i_{\text{N}}$

$R_{\text{th}} = R_{\text{N}}$

$v_{\text{th}} = R_{\text{th}}\, i_{\text{N}}$



Thévenin and Norton: the same line

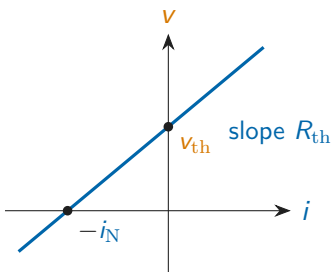
$v = R_{\text{th}}\, i + v_{\text{th}}$

$v = R_{\text{N}}\, i + R_{\text{N}}\, i_{\text{N}}$

$R_{\text{th}} = R_{\text{N}}$

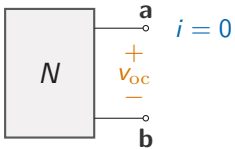
$v_{\text{th}} = R_{\text{th}}\, i_{\text{N}}$

The source transformation (previously)



Computing v_{th}, i_{N} and R_{th}

1. Open circuit

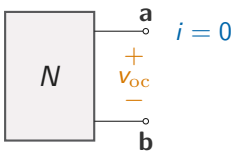


$i = 0$

$i = 0 \;\Rightarrow\; v_{\text{oc}} = v_{\text{th}}$

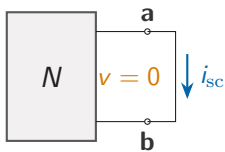
Computing v_{th}, i_{N} and R_{th}

1. Open circuit



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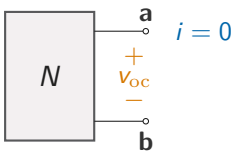
2. Short circuit



$v = 0 \;\Rightarrow\; i_{\text{sc}} = i_{\text{N}}$

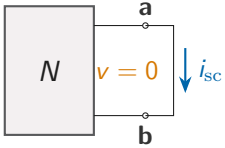
Computing v_{th}, i_{N} and R_{th}

1. Open circuit



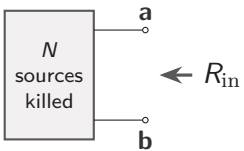
$i = 0 \;\Rightarrow\; v_{\text{oc}} = v_{\text{th}}$

2. Short circuit



$v = 0 \;\Rightarrow\; i_{\text{sc}} = i_{\text{N}}$

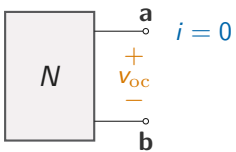
3. R_{th}



$R_{\text{th}} = \dfrac{v_{\text{oc}}}{i_{\text{sc}}}$ or $R_{\text{th}} = R_{\text{in}}$

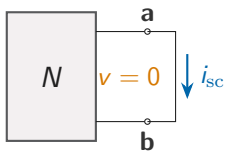
Computing v_{th}, i_{N} and R_{th}

1. Open circuit



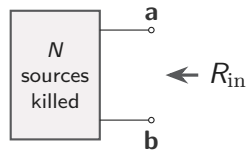
$i = 0 \;\Rightarrow\; v_{\text{oc}} = v_{\text{th}}$

2. Short circuit



$v = 0 \;\Rightarrow\; i_{\text{sc}} = i_{\text{N}}$

3. R_{th}

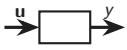


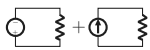
$$R_{\text{th}} = \frac{v_{\text{oc}}}{i_{\text{sc}}} \quad \text{or} \quad R_{\text{th}} = R_{\text{in}}$$

Killing the independent sources: $v_{\text{th}} = 0$, $i_{\text{N}} = 0$, only R_{th} is left

Roadmap











Input resistance ✓

a test source ✓
a dependent source ✓

Linearity ✓

$y = \mathbf{k}\,\mathbf{u}$ ✓
the implication ✓

Superposition ✓

one at a time ✓
three experiments ✓

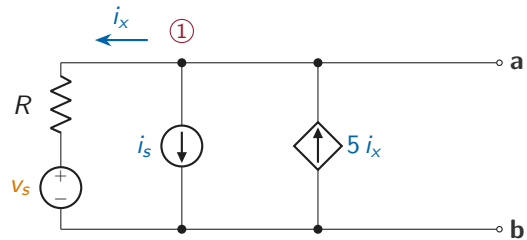
Thévenin, Norton 4

two theorems ✓
open, short ✓
negative R_{th}
two examples

Maximum power 5

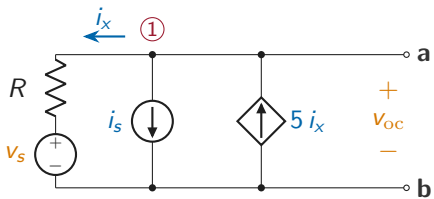
the power P_L
$R_L = R_{\text{th}}$
two examples

Example 13.5: Thévenin and Norton equivalents



$v_{\text{th}} = ?$ $\quad i_{\text{N}} = ?$ $\quad R_{\text{th}} = ?$

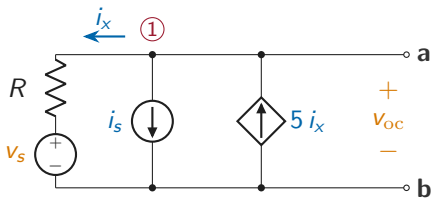
Example 13.5: step 1, open circuit



KCL at node 1:

$$i_x + i_s - 5\,i_x = 0 \;\Rightarrow\; i_x = \tfrac{1}{4}\,i_s$$

Example 13.5: step 1, open circuit



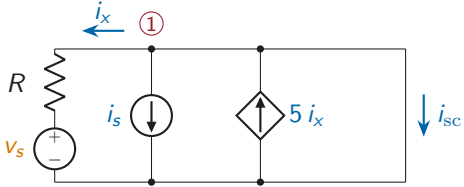
KCL at node 1:

$$i_x + i_s - 5\,i_x = 0 \;\Rightarrow\; i_x = \tfrac{1}{4}\,i_s$$

$$v_{oc} = R\,i_x + v_s$$

$$\boxed{v_{th} = \tfrac{R}{4}\,i_s + v_s}$$

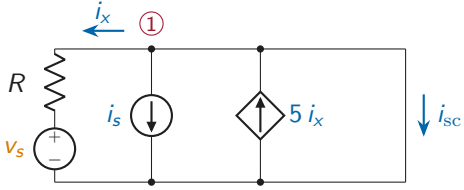
Example 13.5: step 2, short circuit



Node 1 at $0\,\mathrm{V}$:

$$R\,i_x + v_s = 0 \;\Rightarrow\; i_x = -v_s/R$$

Example 13.5: step 2, short circuit



Node 1 at $0\,\mathrm{V}$:

$$R\,i_x + v_s = 0 \;\Rightarrow\; i_x = -v_s/R$$

KCL at node 1:

$$i_x + i_s - 5\,i_x + i_{sc} = 0$$

$$i_N = i_{sc} = -i_s - \frac{4}{R}\,v_s$$

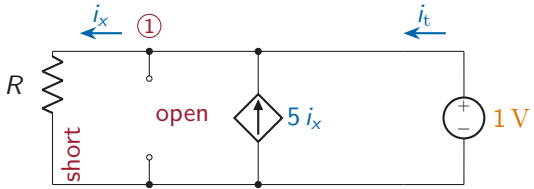
Example 13.5: step 3, R_{th}

$$R_{\mathrm{th}} = \frac{v_{\mathrm{th}}}{i_{\mathrm{N}}} = \frac{\frac{R}{4}\, i_s + v_s}{-\left(i_s + \frac{4}{R}\, v_s\right)} = -\frac{R}{4}$$

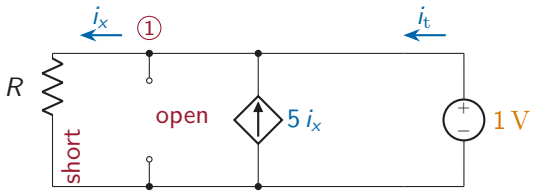
Negative: the dependent source

Example 13.5: R_{th} another way

Kill v_s and i_s; a $1\,\text{V}$ test source



Example 13.5: R_{th} another way

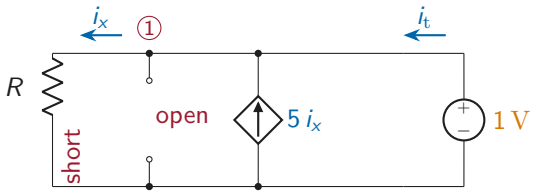


Kill v_s and i_s; a $1\,\text{V}$ test source

$$i_x = \frac{1}{R}$$

KCL: $i_x - 5\,i_x - i_t = 0 \;\Rightarrow\; i_t = -\frac{4}{R}$

Example 13.5: R_{th} another way



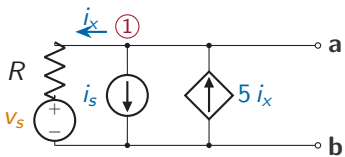
Kill v_s and i_s; a 1 V test source

$$i_x = \frac{1}{R}$$

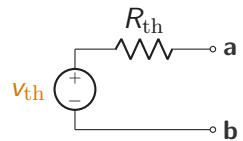
KCL: $i_x - 5\,i_x - i_t = 0 \;\Rightarrow\; i_t = -\frac{4}{R}$

$$R_{\text{th}} = R_{\text{in}} = \frac{1\,\text{V}}{i_t} = -\frac{R}{4} \quad \checkmark$$

Example 13.5: the equivalents



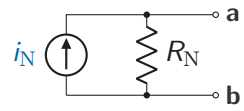
$\equiv$



$$v_{\text{th}} = \frac{R}{4}\, i_s + v_s$$

$$R_{\text{th}} = -\frac{R}{4}$$

$\equiv$

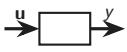


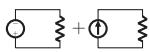
$$i_{\text{N}} = -i_s - \frac{4}{R}\, v_s$$

$$R_{\text{N}} = -\frac{R}{4}$$

Roadmap











Input resistance ✓

a test source ✓
a dependent source ✓

Linearity ✓

$y = \mathbf{k}\,\mathbf{u}$ ✓
the implication ✓

Superposition ✓

one at a time ✓
three experiments ✓

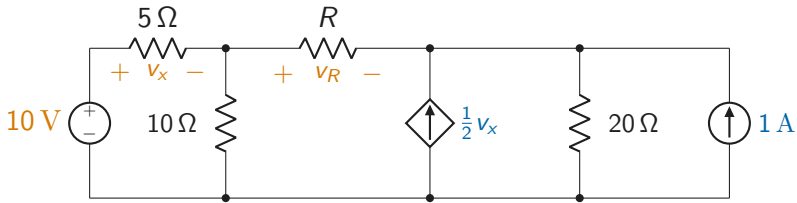
Thévenin, Norton 4

two theorems ✓
open, short ✓
negative R_{th} ✓
two examples

Maximum power 5

the power P_L
$R_L = R_{\text{th}}$
two examples

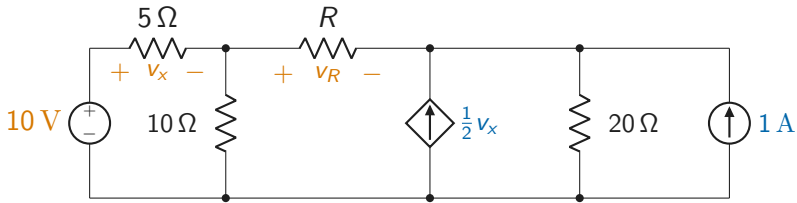
Example 13.6: a load to design



The load: R; the rest of the circuit: a one-port

Thévenin equivalent seen by R?

Example 13.6: a load to design

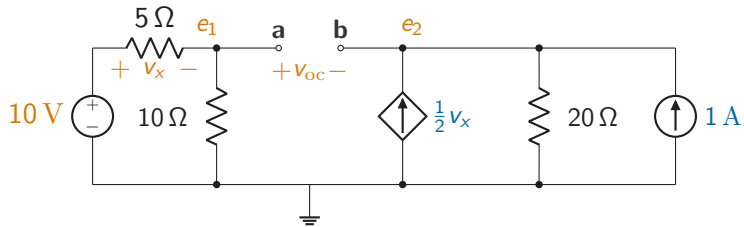


The load: R; the rest of the circuit: a one-port

Thévenin equivalent seen by R?

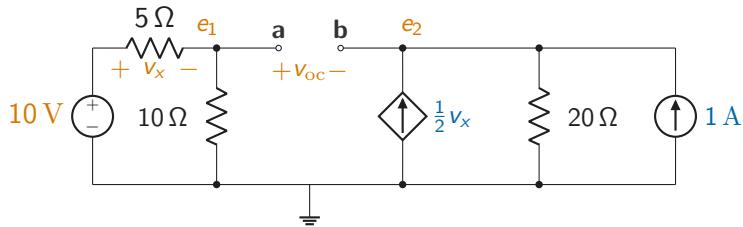
$R = ?$ for $v_R = -7\,\text{V}$

Example 13.6: open circuit



Node 1: $\dfrac{e_1 - 10}{5} + \dfrac{e_1}{10} = 0 \;\Rightarrow\; e_1 = \frac{20}{3}\,\text{V}$

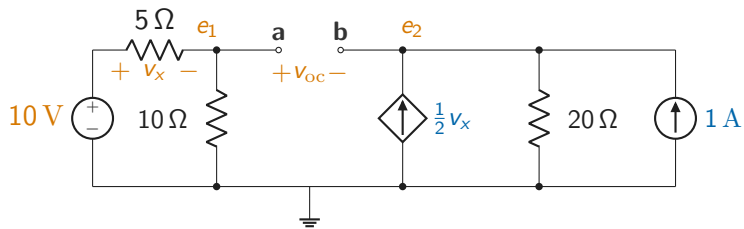
Example 13.6: open circuit



Node 1: $\dfrac{e_1 - 10}{5} + \dfrac{e_1}{10} = 0 \;\Rightarrow\; e_1 = \frac{20}{3}\,\text{V}$

$v_x = 10 - e_1 = \frac{10}{3}\,\text{V}$

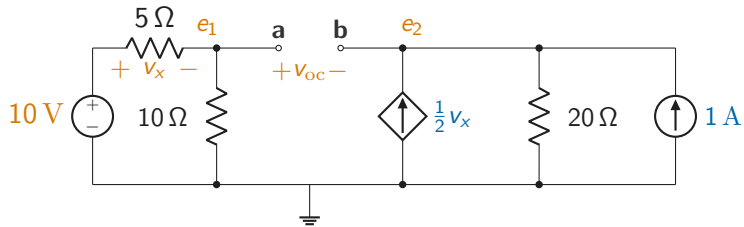
Example 13.6: open circuit



Node 1: $\dfrac{e_1 - 10}{5} + \dfrac{e_1}{10} = 0 \Rightarrow e_1 = \frac{20}{3}\,\text{V}$ **Node 2:** $-\frac{1}{2}v_x + \dfrac{e_2}{20} - 1 = 0 \Rightarrow$ $e_2 = \frac{160}{3}\,\text{V}$

$v_x = 10 - e_1 = \frac{10}{3}\,\text{V}$

Example 13.6: open circuit

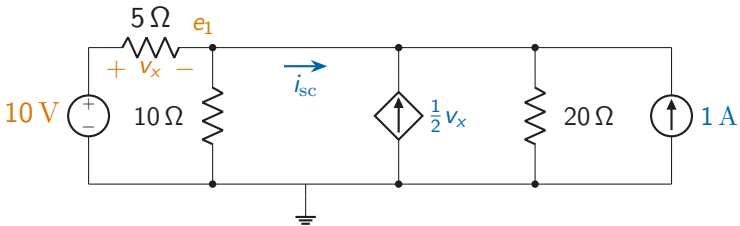


Node 1: $\dfrac{e_1 - 10}{5} + \dfrac{e_1}{10} = 0 \Rightarrow e_1 = \frac{20}{3}\,\text{V}$ **Node 2:** $-\frac{1}{2}v_x + \dfrac{e_2}{20} - 1 = 0 \Rightarrow$ $e_2 = \frac{160}{3}\,\text{V}$

$v_x = 10 - e_1 = \frac{10}{3}\,\text{V}$

$$v_{\text{th}} = e_1 - e_2 = -\frac{140}{3}\,\text{V}$$

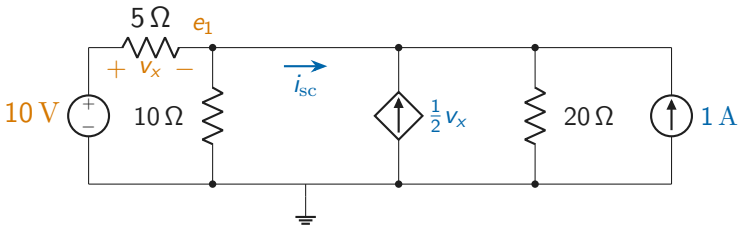
Example 13.6: short circuit



One node e_1:

$$\frac{e_1 - 10}{5} + \frac{e_1}{10} - \tfrac{1}{2}v_x + \frac{e_1}{20} - 1 = 0$$

Example 13.6: short circuit

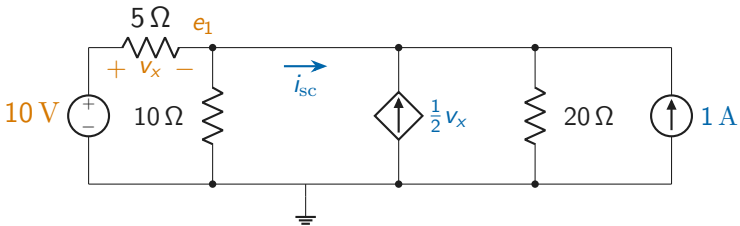


One node e_1:

$$\frac{e_1 - 10}{5} + \frac{e_1}{10} - \tfrac{1}{2}v_x + \frac{e_1}{20} - 1 = 0$$

Plug $v_x = 10 - e_1$: $\frac{17}{20}\,e_1 = 8 \;\Rightarrow\; e_1 = \frac{160}{17}\,\text{V}$

Example 13.6: short circuit



One node e_1:

$$\frac{e_1 - 10}{5} + \frac{e_1}{10} - \tfrac{1}{2}v_x + \frac{e_1}{20} - 1 = 0$$

Plug $v_x = 10 - e_1$: $\frac{17}{20}e_1 = 8 \Rightarrow e_1 = \frac{160}{17}\,\mathrm{V}$

KCL, left of the short:

$$\frac{e_1 - 10}{5} + \frac{e_1}{10} + i_{sc} = 0$$

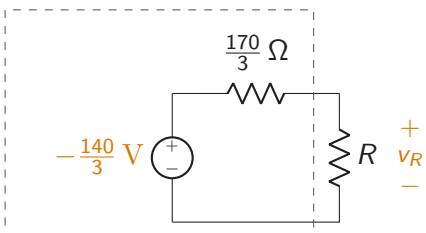
$$i_N = i_{sc} = -\tfrac{14}{17}\,\mathrm{A}$$

Example 13.6: finding R

$$R_{\text{th}} = \frac{v_{\text{th}}}{i_{\text{N}}} = \frac{-140/3}{-14/17} = \frac{170}{3}\,\Omega$$

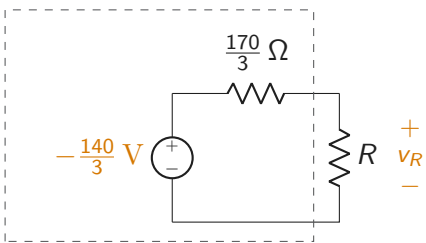
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Example 13.6: finding R

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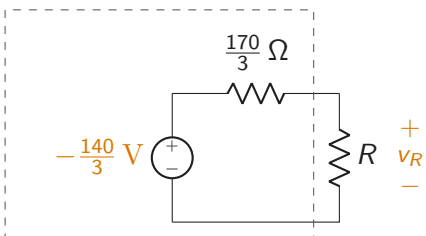


Voltage divider:

$$v_R = \frac{R}{R + \frac{170}{3}}\left(-\frac{140}{3}\right) = -7$$

Example 13.6: finding R

$$R_{\text{th}} = \frac{v_{\text{th}}}{i_{\text{N}}} = \frac{-140/3}{-14/17} = \frac{170}{3}\,\Omega$$



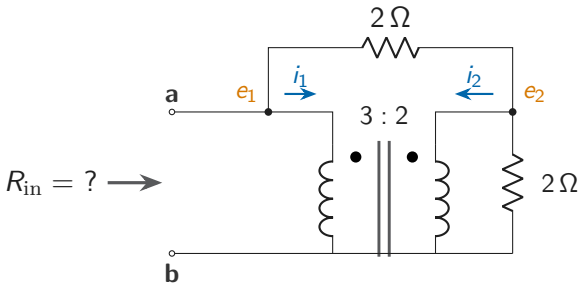
Voltage divider:

$$v_R = \frac{R}{R + \frac{170}{3}}\left(-\frac{140}{3}\right) = -7$$

$$140\,R = 21\,R + 1190$$

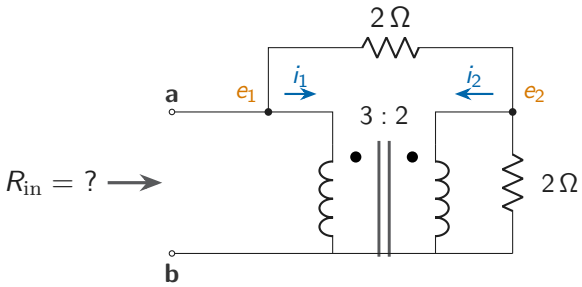
$$\boxed{R = 10\,\Omega}$$

Example 13.7: an ideal transformer



Port: a–b

Example 13.7: an ideal transformer

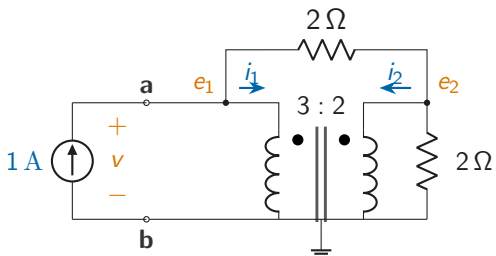


Port: a–b

No independent source inside:
$R_{\text{th}} = R_{\text{in}}$

Ideal transformer 3 : 2:
$\frac{e_1}{3} = \frac{e_2}{2}, \quad 3\,i_1 + 2\,i_2 = 0$

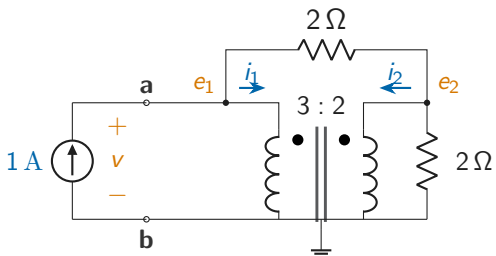
Example 13.7: a $1\,\mathrm{A}$ test source



Node 1: $-1 + i_1 + \dfrac{e_1 - e_2}{2} = 0$

Node 2: $i_2 + \dfrac{e_2}{2} + \dfrac{e_2 - e_1}{2} = 0$

Example 13.7: a $1\,\mathrm{A}$ test source



Node 1: $-1 + i_1 + \dfrac{e_1 - e_2}{2} = 0$

Node 2: $i_2 + \dfrac{e_2}{2} + \dfrac{e_2 - e_1}{2} = 0$

Plug $e_2 = \frac{2}{3}\, e_1$, $i_2 = -\frac{3}{2}\, i_1$:

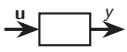
Node 2: $i_1 = \frac{1}{9}\, e_1$

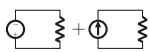
Node 1: $\frac{5}{18}\, e_1 = 1 \;\Rightarrow\; e_1 = \frac{18}{5}\,\mathrm{V}$

$$R_{\mathrm{in}} = \frac{v}{1\,\mathrm{A}} = \frac{e_1}{1\,\mathrm{A}} = \frac{18}{5}\,\Omega$$

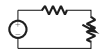
Roadmap











Input resistance ✓

a test source ✓
a dependent source ✓

Linearity ✓

$y = \mathbf{k}\,\mathbf{u}$ ✓
the implication ✓

Superposition ✓

one at a time ✓
three experiments ✓

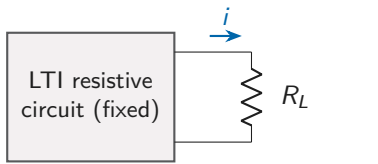
Thévenin, Norton ✓

two theorems ✓
open, short ✓
negative R_{th} ✓
two examples ✓

Maximum power 5

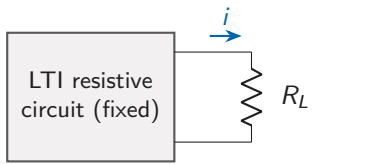
the power P_L
$R_L = R_{\text{th}}$
two examples

Motivation: the most power



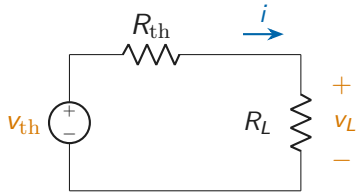
- An amplifier drives a speaker; an antenna feeds a receiver
- The circuit is fixed; we choose the load R_L

Motivation: the most power



- An amplifier drives a speaker; an antenna feeds a receiver
- The circuit is fixed; we choose the load R_L
- $R_L = ?$ for the maximum power to the load

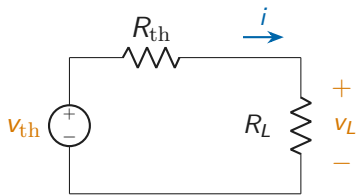
The power delivered to the load



$$i = \frac{v_{\text{th}}}{R_{\text{th}} + R_L}$$

v_{th}, R_{th}: fixed

The power delivered to the load



v_{th}, R_{th}: fixed

$$i = \frac{v_{\text{th}}}{R_{\text{th}} + R_L}$$

$$P_L = R_L\, i^2 = \frac{R_L\, {v_{\text{th}}}^2}{(R_{\text{th}} + R_L)^2}$$

P_L: a function of R_L only

Two extremes

$R_L = 0$ (short): $v_L = 0, \quad P_L = 0$

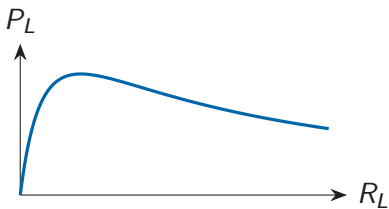
$R_L \to \infty$ (open): $i = 0, \quad P_L \to 0$

Two extremes

$R_L = 0$ (short): $v_L = 0, \quad P_L = 0$

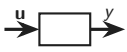
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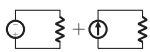
The maximum is in between



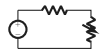
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Thévenin, Norton ✓

two theorems ✓
open, short ✓
negative R_{th} ✓
two examples ✓

Maximum power 5

the power P_L ✓
$R_L = R_{\text{th}}$
two examples

Maximum power transfer: $R_L = R_{\text{th}}$

$$\frac{dP_L}{dR_L} = v_{\text{th}}{}^2 \frac{(R_{\text{th}} + R_L) - 2R_L}{(R_{\text{th}} + R_L)^3}$$

$$= v_{\text{th}}{}^2 \frac{R_{\text{th}} - R_L}{(R_{\text{th}} + R_L)^3}$$

Maximum power transfer: $R_L = R_{\text{th}}$

$$\frac{dP_L}{dR_L} = v_{\text{th}}{}^2 \frac{(R_{\text{th}} + R_L) - 2R_L}{(R_{\text{th}} + R_L)^3}$$

$$= v_{\text{th}}{}^2 \frac{R_{\text{th}} - R_L}{(R_{\text{th}} + R_L)^3}$$

> 0 for $R_L < R_{\text{th}}$, < 0 for $R_L > R_{\text{th}}$

Maximum power transfer: $R_L = R_{\text{th}}$

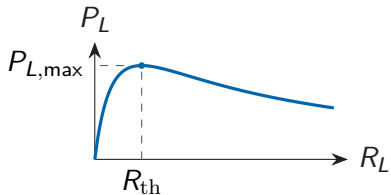
$$\frac{dP_L}{dR_L} = {v_{\text{th}}}^2 \frac{(R_{\text{th}} + R_L) - 2R_L}{(R_{\text{th}} + R_L)^3}$$

$$= {v_{\text{th}}}^2 \frac{R_{\text{th}} - R_L}{(R_{\text{th}} + R_L)^3}$$

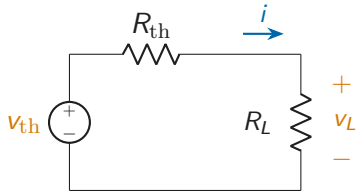
> 0 for $R_L < R_{\text{th}}$, < 0 for $R_L > R_{\text{th}}$

$$R_L = R_{\text{th}}$$

$$P_{L,\text{max}} = \frac{{v_{\text{th}}}^2}{4R_{\text{th}}}$$



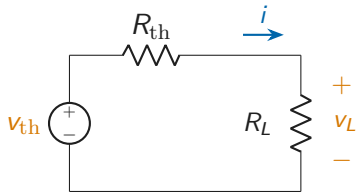
At maximum power



$R_L = R_{\text{th}}$

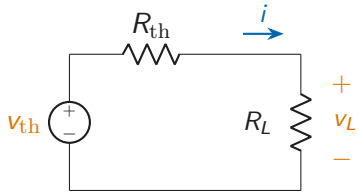
- $v_L = \frac{1}{2}\, v_{\text{th}}, \quad i = \dfrac{v_{\text{th}}}{2R_{\text{th}}}$

At maximum power



- $v_L = \frac{1}{2}\, v_{\text{th}}, \quad i = \dfrac{v_{\text{th}}}{2R_{\text{th}}}$
- Norton form: $P_{L,\max} = \frac{1}{4}\, R_{\text{N}}\, {i_{\text{N}}}^2$

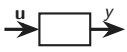
At maximum power

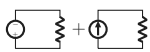


- $v_L = \frac{1}{2}\, v_{\text{th}}, \quad i = \dfrac{v_{\text{th}}}{2R_{\text{th}}}$
- Norton form: $P_{L,\text{max}} = \frac{1}{4}\, R_{\text{N}}\, {i_{\text{N}}}^2$
- In the equivalent, half of the source power reaches the load

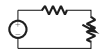
Roadmap











Input resistance ✓

a test source ✓
a dependent source ✓

Linearity ✓

$y = \mathbf{k}\,\mathbf{u}$ ✓
the implication ✓

Superposition ✓

one at a time ✓
three experiments ✓

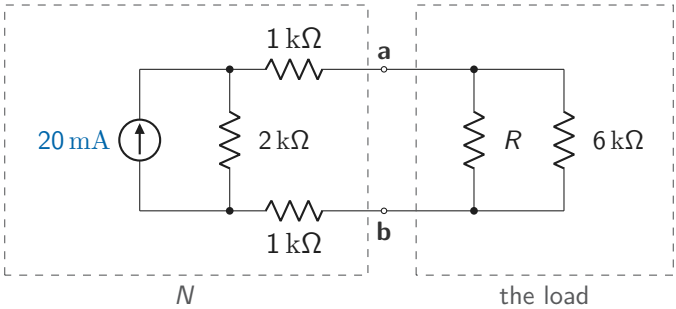
Thévenin, Norton ✓

two theorems ✓
open, short ✓
negative R_{th} ✓
two examples ✓

Maximum power 5

the power P_L ✓
$R_L = R_{\text{th}}$ ✓
two examples

Example 13.8: maximum power

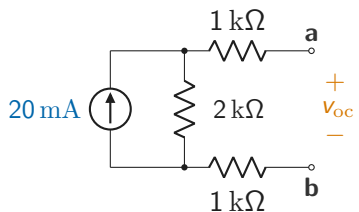


$R = ?$ for the maximum power to the load

$P_{L,\text{max}} = ?$

Example 13.8: the Thévenin equivalent of N

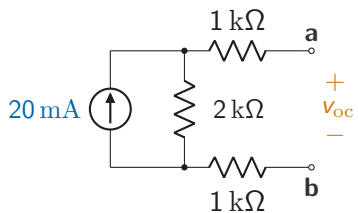
Open circuit



$$v_{th} = v_{oc} = 20\,\text{mA} \times 2\,\text{k}\Omega = 40\,\text{V}$$

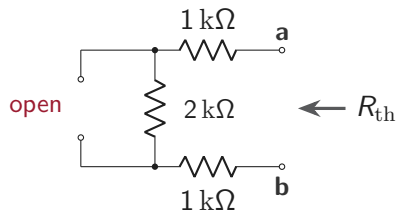
Example 13.8: the Thévenin equivalent of N

Open circuit



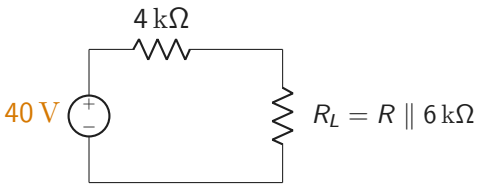
$$v_{th} = v_{oc} = 20\,\text{mA} \times 2\,\text{k}\Omega = 40\,\text{V}$$

Kill the source



$$R_{th} = 1 + 2 + 1 = 4\,\text{k}\Omega$$

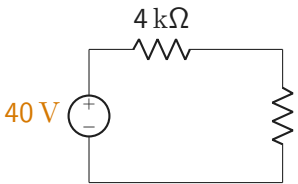
Example 13.8: R and the maximum power



$R_L = R_{\mathrm{th}} = 4\,\mathrm{k\Omega}$:

$$\frac{1}{R} + \frac{1}{6} = \frac{1}{4} \;\Rightarrow\; R = 12\,\mathrm{k\Omega}$$

Example 13.8: R and the maximum power



$R_L = R \parallel 6\,\mathrm{k\Omega}$

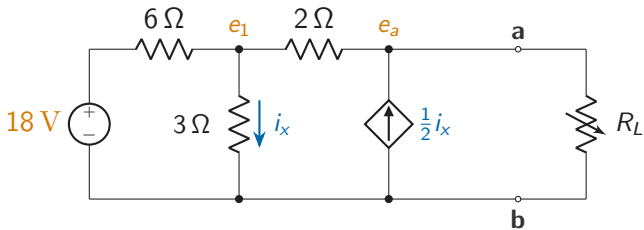
$R_L = R_{\mathrm{th}} = 4\,\mathrm{k\Omega}$:

$$\frac{1}{R} + \frac{1}{6} = \frac{1}{4} \;\Rightarrow\; R = 12\,\mathrm{k\Omega}$$

$$P_{L,\max} = 4\,\mathrm{k\Omega}\left(\frac{40\,\mathrm{V}}{8\,\mathrm{k\Omega}}\right)^2 = 4\,\mathrm{k\Omega}\,(5\,\mathrm{mA})^2$$

$$\boxed{P_{L,\max} = 0.1\,\mathrm{W}}$$

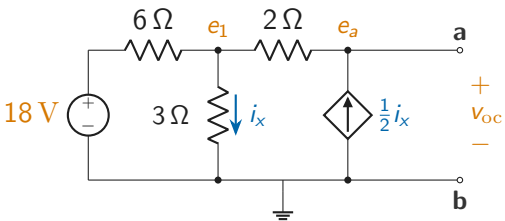
Example 13.9: Thévenin, Norton, maximum power



Thévenin and Norton equivalents seen by R_L?

$R_L = ?$ for the maximum power; $P_{L,\max} = ?$

Example 13.9: open circuit



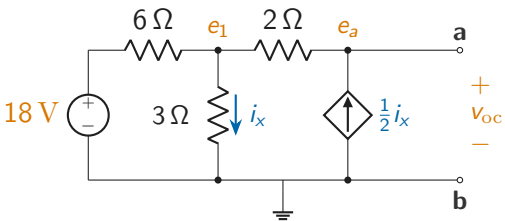
Plug $i_x = e_1/3$ first.

Node 1: $\dfrac{e_1 - 18}{6} + \dfrac{e_1}{3} + \dfrac{e_1 - e_a}{2} = 0$

Node a: $\dfrac{e_a - e_1}{2} - \dfrac{e_1}{6} = 0 \Rightarrow$

$e_a = \frac{4}{3}\, e_1$

Example 13.9: open circuit



Plug $i_x = e_1/3$ first.

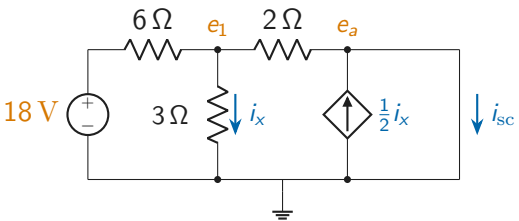
Node 1: $\dfrac{e_1 - 18}{6} + \dfrac{e_1}{3} + \dfrac{e_1 - e_a}{2} = 0$

Node a: $\dfrac{e_a - e_1}{2} - \dfrac{e_1}{6} = 0 \Rightarrow$

$e_a = \frac{4}{3}\, e_1$

$e_1 = 9\,\mathrm{V}$, $\boxed{v_{\mathrm{th}} = v_{\mathrm{oc}} = e_a = 12\,\mathrm{V}}$

Example 13.9: short circuit



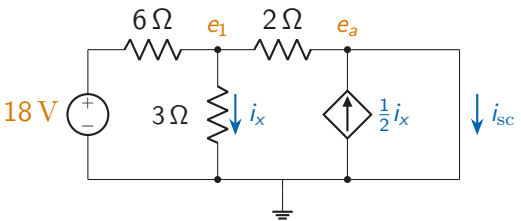
$e_a = 0$:

Node 1: $\dfrac{e_1 - 18}{6} + \dfrac{e_1}{3} + \dfrac{e_1}{2} = 0 \Rightarrow$

$e_1 = 3\,\text{V}$

$i_x = 1\,\text{A}$

Example 13.9: short circuit



$e_a = 0$:

Node 1: $\dfrac{e_1 - 18}{6} + \dfrac{e_1}{3} + \dfrac{e_1}{2} = 0 \Rightarrow$

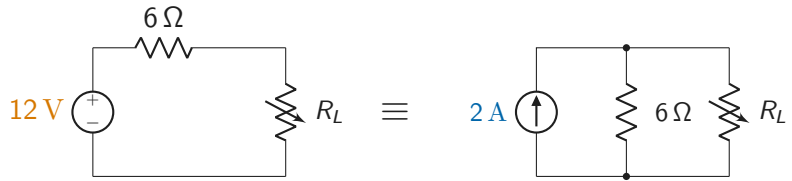
$e_1 = 3\,\text{V}$

$i_x = 1\,\text{A}$

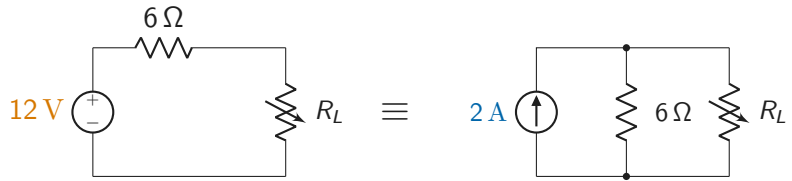
$i_N = i_{sc} = \dfrac{e_1}{2} + \frac{1}{2} i_x = 2\,\text{A}$

$$R_{th} = R_N = \frac{v_{th}}{i_N} = 6\,\Omega$$

Example 13.9: the equivalents and the maximum power



Example 13.9: the equivalents and the maximum power



$$R_L = R_{\text{th}} = 6\,\Omega$$

$$P_{L,\text{max}} = \frac{{v_{\text{th}}}^2}{4R_{\text{th}}} = \frac{12^2}{24} = \tfrac{1}{4}\,R_{\text{N}}\,{i_{\text{N}}}^2 = \tfrac{1}{4}\cdot 6\cdot 2^2$$

$$\boxed{P_{L,\text{max}} = 6\,\text{W}}$$

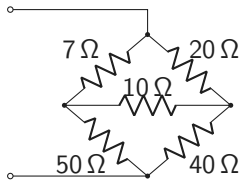
Summary

In the circuit	What to do
resistors, dependent sources	$R_{\text{in}} = v/1\,\text{A}$ with a $1\,\text{A}$ test source
linear circuit	$y = \mathbf{k}\,\mathbf{u}$: $y(\alpha\mathbf{u} + \beta\mathbf{w}) = \alpha y(\mathbf{u}) + \beta y(\mathbf{w})$
several sources	superposition: killed v_s = short, killed i_s = open
independent sources inside	$v_{\text{th}} = v_{\text{oc}}$, $i_{\text{N}} = i_{\text{sc}}$, $R_{\text{th}} = R_{\text{N}} = v_{\text{oc}}/i_{\text{sc}}$ or R_{in} with the sources killed
maximum power to R_L	$R_L = R_{\text{th}}$, $P_{L,\text{max}} = {v_{\text{th}}}^2/(4R_{\text{th}})$

Dependent sources are never killed; R_{th} may be negative

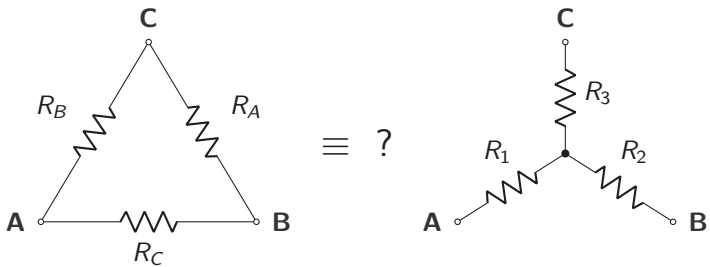
Take-home

The Δ–Y transform

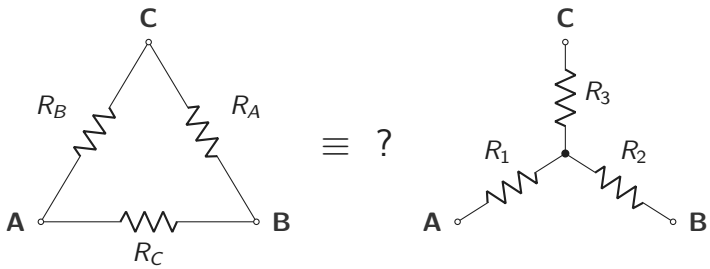


No two resistors in series or in parallel

Three-terminal circuits: Δ and Y

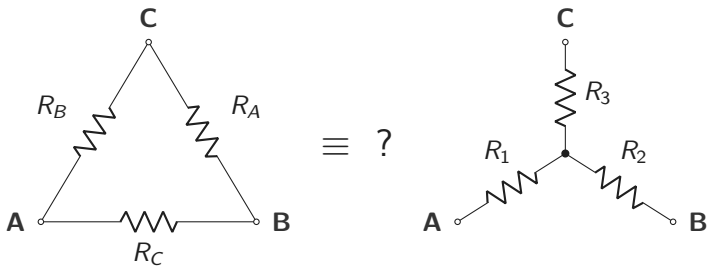


Three-terminal circuits: Δ and Y



Given R_A, R_B, R_C: find R_1, R_2, R_3

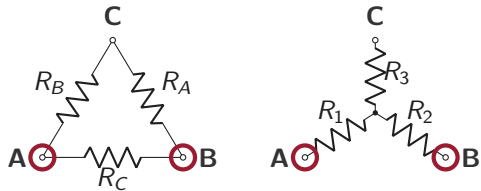
Three-terminal circuits: Δ and Y



Given R_A, R_B, R_C: find R_1, R_2, R_3

The same resistance between every two terminals

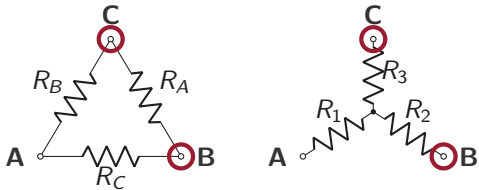
Δ to Y: the resistance between two terminals



Δ Y

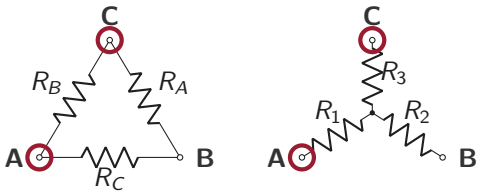
A–B: $$\frac{R_C(R_A + R_B)}{R_A + R_B + R_C} = R_1 + R_2 \quad (1)$$

Δ to Y: the resistance between two terminals



	Δ		Y	
A–B:	$\dfrac{R_C(R_A+R_B)}{R_A+R_B+R_C}$	$=$	R_1+R_2	(1)
B–C:	$\dfrac{R_A(R_B+R_C)}{R_A+R_B+R_C}$	$=$	R_2+R_3	(2)

Δ to Y: the resistance between two terminals



	Δ		Y	
A–B:	$\dfrac{R_C(R_A+R_B)}{R_A+R_B+R_C}$	$=$	R_1+R_2	(1)
B–C:	$\dfrac{R_A(R_B+R_C)}{R_A+R_B+R_C}$	$=$	R_2+R_3	(2)
C–A:	$\dfrac{R_B(R_C+R_A)}{R_A+R_B+R_C}$	$=$	R_3+R_1	(3)

Δ to Y: the result

$$\frac{1}{2}\left[(1)+(2)+(3)\right]: \quad R_1+R_2+R_3=\frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \tag{4}$$

Δ to Y: the result

$$\tfrac{1}{2}\left[(1)+(2)+(3)\right]: \quad R_1+R_2+R_3=\frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \quad (4)$$

$$R_1=\frac{R_BR_C}{R_A+R_B+R_C} \quad (4)-(2)$$

Δ to Y: the result

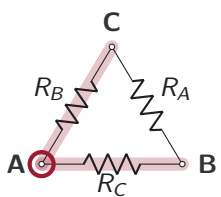
$$\tfrac{1}{2}\big[(1)+(2)+(3)\big]: \quad R_1+R_2+R_3=\frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \quad (4)$$

$$R_1=\frac{R_BR_C}{R_A+R_B+R_C} \quad (4)-(2) \qquad R_2=\frac{R_CR_A}{R_A+R_B+R_C} \qquad R_3=\frac{R_AR_B}{R_A+R_B+R_C}$$

Δ to Y: the result

$$\tfrac{1}{2}[(1)+(2)+(3)]: \quad R_1+R_2+R_3=\frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \quad (4)$$

$$R_1=\frac{R_BR_C}{R_A+R_B+R_C} \quad (4)-(2) \qquad R_2=\frac{R_CR_A}{R_A+R_B+R_C} \qquad R_3=\frac{R_AR_B}{R_A+R_B+R_C}$$



Y resistor at a terminal = product of the two Δ resistors there / sum of the Δ resistors

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_AR_BR_C\,(R_A + R_B + R_C)}{(R_A + R_B + R_C)^2} = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_AR_BR_C\,(R_A + R_B + R_C)}{(R_A + R_B + R_C)^2} = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

$$R_AR_1 = R_BR_2 = R_CR_3 = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_AR_BR_C\,(R_A + R_B + R_C)}{(R_A + R_B + R_C)^2} = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

$$R_AR_1 = R_BR_2 = R_CR_3 = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

$$R_A = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}, \qquad R_B = \frac{\cdots}{R_2}, \qquad R_C = \frac{\cdots}{R_3}$$

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

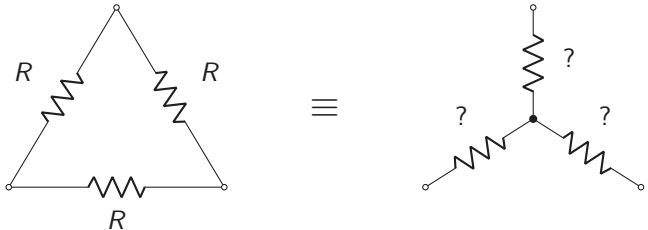
$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_AR_BR_C\,(R_A + R_B + R_C)}{(R_A + R_B + R_C)^2} = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

$$R_AR_1 = R_BR_2 = R_CR_3 = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

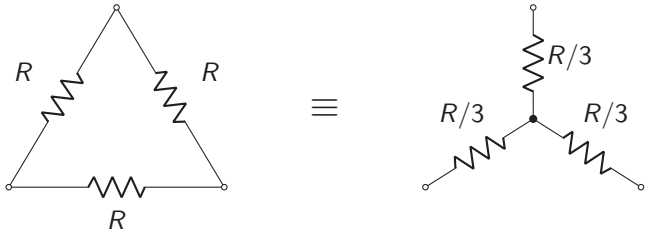
$$R_A = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}, \qquad R_B = \frac{\cdots}{R_2}, \qquad R_C = \frac{\cdots}{R_3}$$

$$\Delta \text{ resistor} = \frac{\text{sum of the products of the Y resistors}}{\text{the Y resistor opposite}}$$

Equal resistors

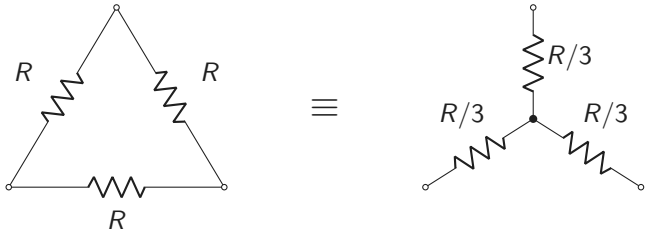


Equal resistors



$$\frac{R \cdot R}{3R} = \frac{R}{3}$$

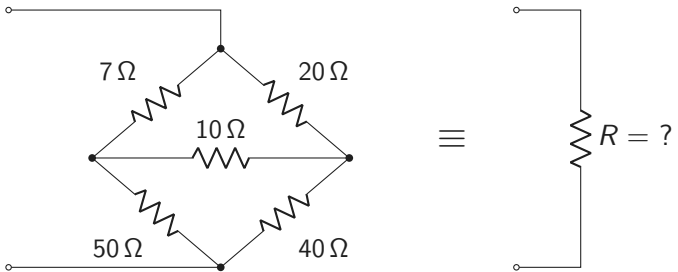
Equal resistors



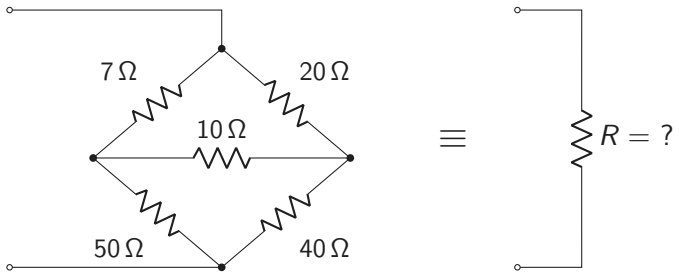
$$\frac{R \cdot R}{3R} = \frac{R}{3}$$

Y with $R \equiv \Delta$ with $3R$

Example 13.10: a bridge

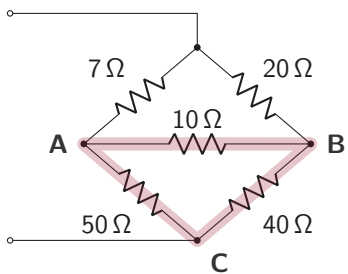


Example 13.10: a bridge



No two resistors in series or in parallel

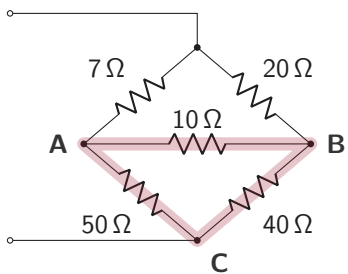
Example 13.10: Δ to Y



Δ A–B–C:

$R_C = 10\,\Omega, \quad R_B = 50\,\Omega, \quad R_A = 40\,\Omega$

Example 13.10: Δ to Y

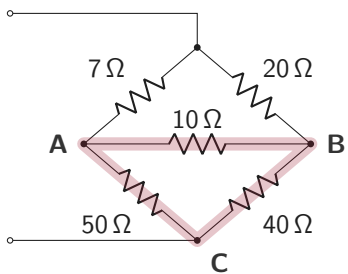


Δ A–B–C:

$R_C = 10\,\Omega, \quad R_B = 50\,\Omega, \quad R_A = 40\,\Omega$

$R_A + R_B + R_C = 100\,\Omega$

Example 13.10: Δ to Y



Δ A–B–C:

$R_C = 10\,\Omega, \quad R_B = 50\,\Omega, \quad R_A = 40\,\Omega$

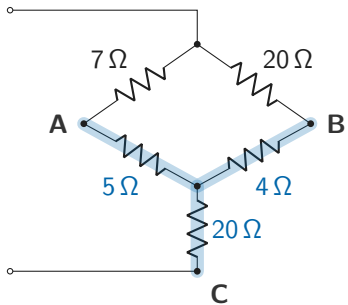
$R_A + R_B + R_C = 100\,\Omega$

$$R_1 = \frac{50 \cdot 10}{100} = 5\,\Omega$$

$$R_2 = \frac{10 \cdot 40}{100} = 4\,\Omega$$

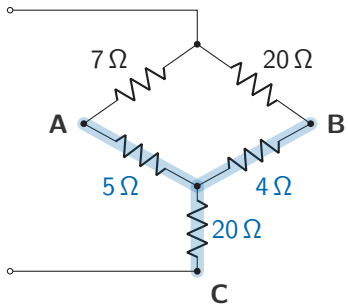
$$R_3 = \frac{40 \cdot 50}{100} = 20\,\Omega$$

Example 13.10: the new circuit



Series: $7 + 5 = 12\,\Omega, \quad 20 + 4 = 24\,\Omega$

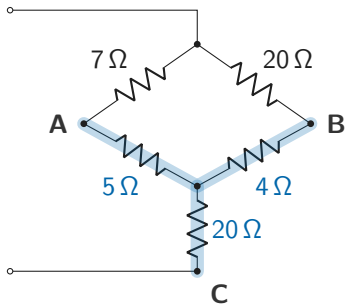
Example 13.10: the new circuit



Series: $7 + 5 = 12\,\Omega, \quad 20 + 4 = 24\,\Omega$

Parallel: $12 \parallel 24 = 8\,\Omega$

Example 13.10: the new circuit



Series: $7 + 5 = 12\,\Omega, \quad 20 + 4 = 24\,\Omega$

Parallel: $12 \parallel 24 = 8\,\Omega$

Series: $8 + 20$

$$R = 28\,\Omega$$